
Assignment Report:

Assignment 4

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In this report, when citing a reference, such as the xv6-book, feel free to use footnotes¹.

Your report should follow the template with the following section structure.

No page limitation

1 Introduction [5']

1. Just as a process is assigned a process id when created, a file is also assigned an inode number upon creation. It has a structure of file name: inode, so the file name matches the inode number. In the file system, files are searched for and accessed based on the inode number, not the file name.
2. I accomplished to learn about relationship between data blocks and index blocks and understand how a given block number refers to the actual data block through a double indirect block by implementing it. Was able to implement double indirect block that enable to handle bigger file.
3. The progress was smooth because school provided the code for direct and single indirect in the bmap function and double indirect was similar to these codes except adding the way to use double indirect block to reference the data.

2 Implementation [3']

- 2.1 **bigfile:** 1). Structure of inode in file.h: As it is mentioned in the assignment4 instruction, I changed the size of addrs from NDIRECT + 1 to NDIRECT + 2 because NDIRECT is changed to 11.
- 2). Structure of dinode and change of definitions in fs.h: As it is mentioned in the instruction, the size of addrs is changed to NDIRECT + 2 and NDIRECT is changed to 11 from 12 which makes 13th element of ip->addrs[] become doubly-indirect block. The size of MAXFILE is added with DNINDIRECT in order to make system able to execute double indirect.
- 3.) bmap: In bmap function, using the given inode and block number, we find location of the block in file system. To make this work, we distinguished between the case of finding a block directly in an inode and the case of finding a block through an indirect block. Using bn/NINDIRECT and bn%NINDIRECT efficiently index a block number into an indirect block.
- 4.) itrunc: In itrunc function, it is responsible for freeing all blocks connected to a given inode. I implemented to find direct and indirect blocks in the inode and released each block that is found. Block was released using bfree function.

2.2 symlink

3 Test [2']

Briefly discuss which part of your implementation helped you pass each test. (Several sentences for each subsection should suffice.)

¹Reference: <https://pdos.csail.mit.edu/6.828/2005/readings/pdp11-40.pdf>

3.1 bigfile: It was important to release the buffer after using it. Without brelse, when I executed, the block stopped showing output after printing out few blocks.

3.2 symlink (if done)

4 Comments [0']

I hope there is more instruction for students on new knowledge students are encountering such as xv6. For me, I followed tutorial in order to connect imported xv6 to visual studio but it didn't work well. I asked the way to solve problem on piazza but there was no answer. It took a long time to solve the problem. I hope the piazza discussion can be more active because after assignment2, the questions on assignment 3 and 4 are not discussed actively.

